SUXAMETHONIUM

High-alert drug

Store in fridge

CHEMICAL NAME 2,2′-[(1,4-dioxobutane-1,4-diyl)bis(oxy)]bis(N,N,N-trimethylethanaminium) dichloride (succinylcholine chloride)

CLASS Depolarising neuromuscular blocking agent

TRADE NAMES Anectine · Scoline · Quelicin · Succinylcholine

THERAPEUTIC USES / INDICATIONS

- Rapid sequence intubation (fast onset, short duration).
- Short procedures needing brief muscle relaxation (e.g. ECT).
- Treatment of laryngospasm.

ADMINISTRATION

- **IV** bolus; **IM** if no IV access.
- Give only after the patient is unconscious — paralyses but does not sedate.
- Ventilation equipment must be ready.
- Store at 2–8 °C (stable 14 days at room temperature).

DOSAGES

- **Adult IV:** 1–1.5 mg/kg.
- **Children IV:** 2 mg/kg (infants), 1–2 mg/kg (children).
- **IM:** 3–4 mg/kg (max 150 mg).
- Pre-treat with atropine in children/repeat doses (bradycardia).

PHARMACOKINETICS

- **Onset:** IV 30–60 s (fasciculations). **Duration:** 5–10 min.
- Rapidly hydrolysed by plasma cholinesterase (pseudocholinesterase).
- Deficiency of this enzyme → prolonged apnoea (hours).

PHARMACODYNAMICS

- Binds **nicotinic acetylcholine receptors** at motor end-plate → persistent depolarisation → fasciculations then flaccid paralysis.
- Not reversed by neostigmine (may worsen).

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Muscle fasciculations then post-operative myalgia.
- Bradycardia (especially second dose, children).
- Raised intraocular and intragastric pressure.

ADVERSE EFFECTS (SERIOUS)

- Hyperkalaemia (K^+ rise ~0.5 mmol/L; massive in burns, crush injury, denervation) → cardiac arrest.
- Malignant hyperthermia trigger.
- Prolonged paralysis (pseudocholinesterase deficiency); "sux apnoea".
- Anaphylaxis.

CONTRAINDICATIONS & CAUTIONS

- Personal/family history of malignant hyperthermia.
- Hyperkalaemia; major burns, crush injury or denervation >24–72 h old; spinal cord injury; prolonged immobility.
- Muscular dystrophies (Duchenne) — hyperkalaemic arrest in children.
- Known pseudocholinesterase deficiency; penetrating eye injury (caution).

KEY INTERACTIONS

- **Neostigmine, organophosphates:** prolong block.
- **Digoxin:** arrhythmias.
- **Aminoglycosides, magnesium:** prolonged block.

NURSING MANAGEMENT

- Ensure sedation/anaesthesia given first; prepare airway and suction.
- Monitor ECG (bradycardia, arrhythmias), SpO_2, K^+.
- Watch for rising $ETCO_2$ and temperature (malignant hyperthermia).
- If apnoea persists >10 min, continue ventilation and sedation; check cholinesterase.
- Check fridge storage and expiry.

PATIENT EDUCATION

- Muscle aches after surgery are common and settle in 1–2 days.
- If you have prolonged paralysis, family members should be tested for enzyme deficiency.

KEY POINTS

- Only depolarising blocker; fastest onset (30–60 s).
- Risks: hyperkalaemia, malignant hyperthermia, bradycardia.
- Metabolised by plasma cholinesterase.
- Avoid in burns/crush >24 h, muscular dystrophy.

SOURCES CHECKED **[87]** Suxamethonium Chloride 50mg/ml Solution for Injection (UK SmPC) **[88]** Succinylcholine Chloride, IM, IV (US label) **[89]** Succinylcholine Chloride, IM, IV (US label, Fresenius Kabi USA LLC)

Full references p. 432–457